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2. 找級數 $\sum_{n=1}^{\infty} \ln \frac{n+1}{n}$ 的 n 項部分和式子並依此求級數和。	2
3. $\sum_{n=1}^{\infty} \frac{(1-\frac{3}{n})^{n^2} e^n}{3n^3}$ 試判級數收斂或發散。	3
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1a. $\int_0^1 \frac{1}{x^{\frac{1}{3}}} \, dx$

定積分化簡	
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1b. $\int_{-\infty}^{\infty} \frac{2x}{(x^2+4)^{\frac{1}{3}}} dx$

定積分化簡

瑕積分定義

2. 找級數 $\sum_{n=1}^{\infty} \ln \frac{n+1}{n}$ 的 n 項部分和式子並依此求級數和。

部分和 S_n 與極限

3. $\sum_{n=1}^{\infty} \frac{(1-\frac{3}{n})^{n^2} e^n}{3n^3}$ 試判級數收斂或發散。

(Root Test / Ratio Test)

Python 程式碼驗證 (SymPy)

```
1 import sympy
2
3 n = sympy.symbols('n', integer=True, positive=True)
4 a_n = ((1 - 3/n)**(n**2) * sympy.exp(n)) / (3 * n**3)
5
6 # 使用 Root Test 進行極限判斷
7 root_limit = sympy.limit(a_n**(1/n), n, sympy.oo)
8
9 print(f"Root Test Limit: {root_limit}")
10 if root_limit < 1:
11     print("Convergent")
12 else:
13     print("Divergent or Inconclusive")
```

4. $\int \tan^2 x \sec^3 x \, dx$

$$\begin{aligned} &\int \tan^2 x \sec^3 x \, dx \\ &= \int (\sec^2 x - 1) \sec^3 x \, dx \\ &= \int (\sec^5 x - \sec^3 x) \, dx \\ &= \int \sec^5 x \, dx - \int \sec^3 x \, dx \quad \text{分開討論 ①, ②} \end{aligned}$$

$\int \sec^3 x \, dx$	$\int \sec^5 x \, dx$
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$$I_3 = \int \sec^3 x \, dx$$

$$= \sec x \tan x - \int \sec x \tan^2 x \, dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx$$

$$= \sec x \tan x - I_3 + \int \sec x \, dx$$

$$2I_3 = \sec x \tan x + \ln|\sec x + \tan x|$$

$$I_3 = \frac{1}{2}(\sec x \tan x + \ln|\sec x + \tan x|)$$

$$I_5 = \int \sec^5 x \, dx$$

$$= \sec^3 x \tan x - \int \tan x \cdot 3 \sec^2 x (\sec x \tan x) \, dx$$

$$= \sec^3 x \tan x - 3 \int \sec^3 x (\sec^2 x - 1) \, dx$$

$$= \sec^3 x \tan x - 3I_5 + 3I_3$$

$$4I_5 = \sec^3 x \tan x + 3I_3$$

$$I_5 = \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} I_3$$

$$= \left(\frac{1}{4} \sec^3 x \tan x + \frac{3}{4} I_3 \right) - I_3$$

$$= \frac{1}{4} \sec^3 x \tan x - \frac{1}{4} I_3$$

$$= \frac{1}{4} \sec^3 x \tan x - \frac{1}{4} \left[\frac{1}{2} (\sec x \tan x + \ln|\sec x + \tan x|) \right] + C$$

$$= \frac{1}{4} \sec^3 x \tan x - \frac{1}{8} \sec x \tan x - \frac{1}{8} \ln|\sec x + \tan x| + C$$

5. $\int \cos^4 x \, dx$

$$\int \cos^4 x \, dx$$

$$= \int (\cos^2 x)^2 \, dx$$

$$= \int \left(\frac{1 + \cos 2x}{2} \right)^2 \, dx$$

$$= \frac{1}{4} \int (1 + 2 \cos 2x + \cos^2 2x) \, dx$$

降次討論： $\cos^2 2x$

$$\cos^2 2x = \frac{1 + \cos 4x}{2}$$

$$\int \cos^2 2x \, dx = \int \frac{1 + \cos 4x}{2} \, dx$$

$$= \frac{1}{2} x + \frac{1}{8} \sin 4x$$

$$\begin{aligned}
&= \frac{1}{4} \left[\int 1 \, dx + 2 \int \cos 2x \, dx + \int \cos^2 2x \, dx \right] \\
&= \frac{1}{4} \left[x + \sin 2x + \left(\frac{1}{2}x + \frac{1}{8} \sin 4x \right) \right] + C \\
&= \frac{1}{4} \left[\frac{3}{2}x + \sin 2x + \frac{1}{8} \sin 4x \right] + C \\
&= \frac{3}{8}x + \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + C
\end{aligned}$$

6. $\int \frac{16-x^3}{(x-2)^2(x^2+4)} \, dx$

$$\begin{aligned}
\frac{16-x^3}{(x-2)^2(x^2+4)} &= \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx+D}{x^2+4} \\
\Rightarrow 16-x^3 &= (x-2)(x^2+4)A + (x^2+4)B + (x-2)^2(Cx+D) \\
&= (x^3-2x^2+4x-8)A + (x^2+4)B + (x^3-4x^2+4x)C + (x^2-4x+4)D
\end{aligned}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -1 \\ -2 & 1 & -4 & 1 & 0 \\ 4 & 0 & 4 & -4 & 0 \\ -8 & 4 & 0 & 4 & 16 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -1 \\ -2 & 1 & -4 & 1 & 0 \\ 1 & 0 & 1 & -1 & 0 \\ -2 & 1 & 0 & 1 & 4 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & -2 & 1 & -2 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 2 & 1 & 2 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & -2 & 1 & -2 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 4 & 0 & 4 \end{array} \right) \rightarrow$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & -2 & 1 & -2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right)$$

經高斯消去法得：

$$\begin{cases} A = -2 \\ B = 1 \\ C = 1 \\ D = -1 \end{cases}$$

將 A, B, C, D 代回積分計算

$$\int \left(\frac{-2}{x-2} + \frac{1}{(x-2)^2} + \frac{x-1}{x^2+4} \right) dx$$

將分式拆解：

$$\begin{aligned}
&= -2 \int \frac{1}{x-2} dx + \int (x-2)^{-2} dx + \int \frac{x}{x^2+4} dx - \int \frac{1}{x^2+4} dx \\
&= -2 \ln|x-2| + \frac{(x-2)^{-1}}{-1} + \frac{1}{2} \int \frac{d(x^2+4)}{x^2+4} - \int \frac{1}{x^2+2^2} dx \\
&= -2 \ln|x-2| - \frac{1}{x-2} + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C
\end{aligned}$$

Python 程式碼驗證 (SymPy)

```
1 import sympy
2
3 x = sympy.symbols('x')
4 f = (16 - x**3) / ((x - 2)**2 * (x**2 + 4))
5
6 # 計算部分分式
7 partial_fractions = sympy.apart(f)
8 # 計算積分
9 integral_result = sympy.integrate(f, x)
10
11 print(f"Partial Fractions: {partial_fractions}")
12 print(f"Integral: {integral_result}")
```

Output:

```
1 Partial Fractions: (x - 1)/(x**2 + 4) - 2/(x - 2) + (x - 2)**(-2)
2 Integral: -2*log(x - 2) + log(x**2 + 4)/2 - atan(x/2)/2 - 1/(x - 2)
```

7, 8. 三角代換對照： $\int \frac{\sqrt{1-x^2}}{x} dx$ 與 $\int \frac{\sqrt{1+x^2}}{x} dx$

$$\int \frac{\sqrt{1-x^2}}{x} dx$$

$$\int \frac{\sqrt{1+x^2}}{x} dx$$

$$\text{令 } x = \sin \theta, dx = \cos \theta d\theta$$

$$= \int \frac{\cos \theta}{\sin \theta} (\cos \theta d\theta)$$

$$= \int \frac{\cos^2 \theta}{\sin \theta} d\theta$$

$$= \int \frac{1 - \sin^2 \theta}{\sin \theta} d\theta$$

$$= \int (\csc \theta - \sin \theta) d\theta$$

$$= \ln |\csc \theta - \cot \theta| + \cos \theta + C$$

$$= \ln \left| \frac{1 - \sqrt{1-x^2}}{x} \right| + \sqrt{1-x^2} + C$$

$$\text{令 } x = \tan \theta, dx = \sec^2 \theta d\theta$$

$$= \int \frac{\sec \theta}{\tan \theta} (\sec^2 \theta d\theta)$$

$$= \int \frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} \cdot \sec^2 \theta d\theta$$

$$= \int \frac{\sec^2 \theta}{\sin \theta} d\theta$$

$$= \int \frac{1 + \tan^2 \theta}{\sin \theta} d\theta$$

$$= \int (\csc \theta + \sec \theta \tan \theta) d\theta$$

$$= \ln |\csc \theta - \cot \theta| + \sec \theta + C$$

$$= \ln \left| \frac{\sqrt{1+x^2} - 1}{x} \right| + \sqrt{1+x^2} + C$$